

Mixed Reality Solutions for Tremor Disorders: Ergonomic Hand Motion and AR Rehabilitation

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Abstract

This study introduces an innovative mixed reality (MR) system for managing tremor disorders, including essential tremor and Parkinson's disease. The system combines an ergonomic hand motion assistance device that mimics natural hand movements to stabilize tremors with MR-based rehabilitation exercises that dynamically adjust in difficulty. Telehealth capabilities enable remote monitoring and consultations, enhancing accessibility. An iterative design process informed by feedback from medical professionals, patients, and design experts led to significant improvements in motor control, task performance, and user autonomy. This comprehensive MR intervention represents a substantial advancement in tremor disorder treatment, offering a versatile and user-friendly solution.

CCS Concepts

• **Applied computing** → Life and medical sciences; Health informatics; • **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Mixed / augmented reality; Human computer interaction (HCI); Interaction devices; Haptic devices.

Keywords

Augmented Reality, Mixed Reality, Hand Rehabilitation, Parkinson's Disease, Assistive Device, Haptic Feedback

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1 Introduction

Parkinson's disease (PD) affects approximately 1.6% of the older U.S. population, according to Medicare data on 36 million recipients [Bamford and Henderson, 2021]. PD is characterized by tremors, bradykinesia, muscle rigidity, and postural instability, resulting in significant physical impairment and higher mortality risk. Additional complications include autonomic dysfunction, depression,

dementia, and psychosis [Schwalbe et al., 2023]. As the older population grows, there is a pressing need for innovative rehabilitation strategies. Mixed reality (MR) offers promising solutions, delivering personalized interventions and predictive analytics for improved hand dexterity in PD patients. Traditional interventions are generally split between mechanical devices, such as functional gloves, which offer immediate relief by compensating for reduced strength, and virtual rehabilitation, which uses MR for immersive motor skill exercises [Pereira et al., 2020, Mois et al., 2019, Kolsanov et al., 2020]. However, a gap exists in integrating these approaches to combine immediate support with long-term skill development [Dineshkumar et al., 2022, Gu et al., 2022]. This paper introduces a novel MR system to bridge this gap, providing both ergonomic support and immersive exercises.

2 DESIGN PROCESS

2.1 Mock-up of the Device

This study introduces a mixed-reality framework that merges ergonomic hand support with MR-based rehabilitation, integrating physical tremor stabilization with exercises aimed at motor skill improvement [Gonsheer and Lei, 2022]. The system includes an adaptable hand motion assistance device and MR exercises that dynamically adjust in difficulty based on user progress, while telehealth features support remote monitoring and consultations to enhance accessibility [Lei and Li, 2024]. The assistive device prototype, scaled to fit the human hand, features a biomechanically intuitive open-close mechanism that mimics natural hand movements essential for daily tasks like grasping and releasing (see Figure 1). Its default open-hand position aligns with the palm's natural curve, while the closed-hand position simulates a firm grip. In the grabbing cycle, the device remains idle, allowing intuitive grasping, with guided movement to counter tremors. Upon securely holding an object, the servo motor locks the handles, minimizing tremor effects and reducing muscular strain. For the releasing cycle, pressing a button unlocks the servo, allowing the device to follow the natural hand opening for a frictionless release. This seamless transition between positions promotes user independence by reducing the burden of maintaining grip and facilitating easy object release.

2.2 Physical prototype design

To enhance hand dexterity rehabilitation with mixed reality (MR), a 3D-printed prototype was developed to align its gripping mechanism with the human palm's natural curvature. This ergonomic design combines biomechanical precision with advanced technology, supporting stable and precise hand movements. The prototype

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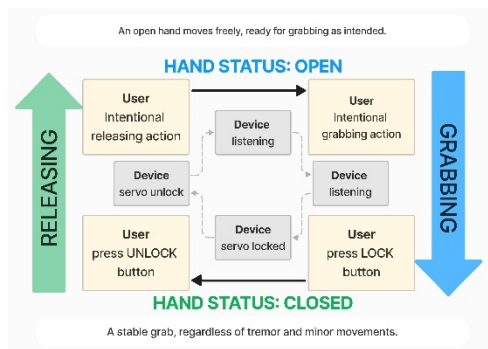


Figure 1: Functional Diagram.

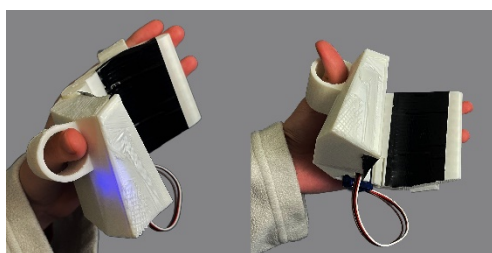


Figure 2: Physical prototype on a human hand.

includes a dual-component system joined by a hinge that replicates the natural thumb-to-finger opposition crucial for gripping. A servo motor embedded in the thumb component functions as both an actuator and stabilizer, providing real-time responsiveness to hand movements (see Figure 2). Powered by an Arduino Nano board with low-energy Bluetooth and a battery in one handle, the device ensures smooth transitions between open and closed hand positions, closely mimicking natural hand dynamics.

2.3 AR System Development

The device integrates seamlessly with a smartphone through a Unity application that uses the phone's camera to provide augmented reality (AR) guidance. Users can point the camera at an object, like a book, and activate the "Grab suggestion" feature, prompting the app to analyze consecutive camera frames and offer optimized grasping techniques. As the user grasps the object, the gyroscope within the device transmits precise movement data, enabling real-time visual feedback on the smartphone screen. This system effectively combines physical interaction with virtual guidance, enhancing task performance by providing intuitive and accurate assistance (see Figure 3).

3 CONCLUSION

This research introduces a novel mixed reality intervention that combines ergonomic hand motion assistance with augmented reality rehabilitation for managing tremor disorders like essential tremor and Parkinson's disease. The integration of physical stabilization and AR-based rehabilitative exercises represents a meaningful advancement in treatment approaches.

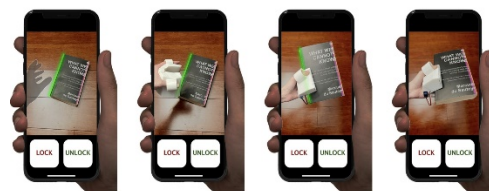


Figure 3: Smartphone AR interface.

The iterative design process, informed by feedback from medical professionals, individuals with tremor disorders, and design experts, has led to a device that improves motor control, task performance, and user autonomy. This innovation addresses both immediate physical needs and long-term rehabilitation, providing a comprehensive solution.

Future work will focus on broader user testing and long-term effectiveness studies to further validate the device's impact. Additionally, refining the user interface and telehealth capabilities will enhance usability and accessibility. This research highlights the potential of mixed reality technologies to offer more effective and user-friendly treatments for tremor disorders, contributing to improved quality of life for those affected.

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